

KATIA LOUNAS

AI & Data Engineer

Paris, France | +33 7 75 71 46 61 | lk.lounas.katia@gmail.com

linkedin.com/in/katia-lounas | github.com/Katia-LOUNAS

PROFESSIONAL SUMMARY

AI and Data Engineer with hands-on experience in LLM integration, prompt engineering, model evaluation, and Python-based machine learning and data pipelines. Dual Master's in Data Science and Intelligent Computer Systems. Experienced in building internal tooling and automated validation workflows for AI-extracted data, benchmarking LLMs across providers, and delivering end-to-end ML projects from data preprocessing to model evaluation. Seeking a full-time AI / Data Engineer role — available immediately.

PROFESSIONAL EXPERIENCE

AI Quality — Apiday, Paris, France

December 2025 – Present

- Automated ESG data validation and extraction workflows using AI, reducing manual review time.
- Built internal tools to validate the accuracy of AI-extracted ESG data across multiple source documents and providers.
- Delivered these workflows with LangChain, the OpenAI API, agentic AI patterns, and Google Cloud Platform.

Internship — Eden AI, Lyon, France

July 2025 – September 2025 (2 months)

- Integrated multiple LLM APIs through Eden AI's unified endpoint and engineered prompts that worked consistently across providers.
- Built an LLM leaderboard to benchmark and compare the models available through Eden AI's API, using Ragas and Opik for evaluation.

Internship — Laboratoire Lyon 1, Lyon, France

October 2024 – February 2025 (4 months)

- Evaluated the Y Social platform for simulating social dynamics and analyzed simulation results using data mining techniques.
- Contributed findings to lab research on LLM-driven social simulations, using LLMs, Y Social, SQLite3, NetworkX, and Matplotlib.

Internship — IA & SD, USTHB

December 2023 – June 2024 (6 months) | Wildfire Spread Prediction

- Preprocessed the SeasFire dataset with xarray to prepare it for wildfire propagation modeling.
- Built and evaluated machine learning models in TensorFlow to predict wildfire spread.

TECHNICAL SKILLS

Programming: Python (primary), Java, C, C++

AI & Machine Learning: Machine Learning, Deep Learning, LLM Evaluation, Prompt Engineering, Agentic AI, Data Mining, Statistics & Probability, Graph Theory & Optimization, Algorithms & Data Structures

AI Frameworks & APIs: LangChain, OpenAI API, Eden AI API, Ragas, Opik, TensorFlow, Scikit-learn

Data & Databases: Pandas, NumPy, xarray, NetworkX, Matplotlib, NLTK, Transformers, Oracle, MySQL, MongoDB, SQLite3

Cloud & DevOps: Google Cloud Platform (GCP), Git, GitLab, CI/CD, Jest

Web Development: JavaScript, Next.js, HTML, CSS, Leaflet

Design Tools: Figma, Adobe Illustrator, Adobe Premiere Pro

Languages: French (native), English (advanced, TOEIC), Kabyle (native)

PROJECTS

EcoTrajet — Carbon Footprint Web Application | Université Lyon 1, 2025

- Built a web application comparing the carbon footprint of transport modes, with route calculation and an interactive map, using Next.js, Leaflet, OSRM, Nominatim, Jest, and GitLab CI/CD.

Information Retrieval System (SRI) | USTHB, 2024

- Built an information retrieval system implementing Boolean, vector-space, probabilistic, and neural models, and evaluated retrieval performance on the LISA collection using NLTK and Transformers.

Data Mining System | USTHB, 2023

- Built a data mining application covering data preprocessing, supervised and unsupervised learning, and model evaluation, using Scikit-learn, Pandas, NumPy, and Matplotlib.

EDUCATION

M.Sc. Data Science & Business Intelligence — EDC Paris, Paris, France

2025 – 2026

Master 1, Computer Science — Université Claude Bernard Lyon 1, Lyon, France

2024 – 2025

M.Sc. Intelligent Computer Systems — USTHB

2022 – 2024 | *Graduated with High Honors*

VOLUNTEER EXPERIENCE

- **ETIC Student Club, ESI (2019 – 2024):** Produced videos and led external relations and event logistics.
- **Micro Club, USTHB (since 2022):** Contributed to IT, communication, and UX/UI design for club projects.

INTERESTS

Hackathons, video editing and production, graphic design, running.